

Fully charged projectiles that are thrown or fired at a nearby shulker will not deflect properly

MC-270852 Projectiles

Status	Resolved / Fixed
Affected	Minecraft 15w31a, 1.20.4, 1.20.5 Pre-Release 3, 1.21, 24w36a
Fixed in	24w37a
Reported	2024-04-16

STEPS TO REPRODUCE

1. Create or open a Creative world where commands can be used.
2. Execute `/summon minecraft:shulker ~ ~ ~ {NoAI:1b}`.
3. Move to a position 1-3 blocks away from the shulker.
4. Fully charge a bow.
5. Fire an arrow at the shulker.
6. Observe that the arrow is almost instantly deflected toward the camera's position instead of getting near the shulker, creating arrow and particle effects in the player's view.
7. Move to a position at least 4.5 blocks away from the shulker.
8. Fully charge the bow and fire another arrow at the shulker.
9. Observe that the arrow gets near the shulker before being deflected.

OBSERVED BEHAVIOUR

When a fully charged arrow is fired at a closed, stationary shulker from approximately 1-3 blocks away, it is deflected almost immediately, before reaching the shulker's hitbox. The deflected arrow travels toward the player's camera position rather than being visibly redirected after approaching the shulker.

Repeatedly firing arrows at this distance rapidly produces arrows and deflection particles around the player's viewpoint, causing significant visual clutter. The arrows do not damage the player in this situation, consistent with MC-125936 and/or MC-214546.

When the same test is performed from approximately 4.5 blocks away or farther, the arrow first travels near the shulker before being deflected as expected. A similar issue can be reproduced with breezes, although the premature deflection is less noticeable.

EXPECTED BEHAVIOUR

A fully charged arrow fired at a nearby closed shulker or breeze should travel toward the entity and reach near its hitbox before being properly deflected away. The deflection should behave consistently at close range as it does from farther away, without redirecting the projectile toward the player's camera or creating excessive visual clutter around the player.

ENVIRONMENT

Minecraft Java Edition Versions: 15w31a, 1.20.4, 1.20.5 Pre-Release 3, 1.21, 24w36a (includes snapshots)

ORIGINAL DESCRIPTION

When the player shoots a projectile at a closed shulker or a breeze from a distance of 1-3 blocks away, the projectile will almost instantly be deflected at the camera's position, never getting near the entity.

Steps to Reproduce:

- Summon a shulker with no AI:

```
/summon minecraft:shulker ~ ~ ~ {NoAI:1b}
```

- Fire an arrow at it from 1-3 blocks away

→ observe result

- Fire an arrow at it from ≥ 4.5 blocks away

→ observe result

Observed Behavior:

When firing it from a closer distance, the arrow will almost instantly be deflected at the camera position, and the player's view will be visually cluttered from all the arrow and particles created. The arrow will also not damage the player due to

MC-125936

Reopened

and/or

MC-214546

Reopened

. When attempting this same issue with breezes, the behavior is less noticable comparatively (shown in the video/s)

Expected Behavior:

The arrow would get near the shulker's hitbox before being deflected properly at close distances just as it does from further away.

Screenshots/Videos:

Reproducing the issue from a third-person perspective:

ShulkerDeflectClose1.mp4

ShulkerDeflectClose2.mp4

Reproducing the issue in first person:

ShulkerDeflectClosePlayerView.mp4

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-270852>